REUBEN NEWTON ADDISON, PhD

Academic Curriculum Vitae

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RESEARCH PROFILE

Assistant Professor of Kinesiology and Director of the Motor Control Lab whose research connects motor behavior, clinical movement outcomes, and data science. Scholarship spans Parkinson's disease, visuomotor adaptation, neurological research, and clinical measurement. Quantitative work includes Bayesian and longitudinal modeling, biomedical machine learning, and reproducible research software. Teaching and mentorship connect human movement science with experimental design and data analysis.

EDUCATION

MS in Analytics | Georgia Institute of Technology | May 2026 | Computational Data Analytics Track

PhD in Kinesiology | Louisiana State University | May 2022 | Motor Behavior

MS in Kinesiology | Memorial University of Newfoundland | May 2017

BS in Psychology | University of Ghana | May 2012

Doctoral dissertation: Visuomotor rotation adaptation and workspace manipulation: A behavioral and cognitive emphasis.

Master's thesis: Cognitive load assessment in computer-based video training for one-hand knot tying.

Undergraduate thesis: Attitudes towards suicide-attempters.

ACADEMIC AND RESEARCH APPOINTMENTS

Assistant Professor of Kinesiology and Director of the Motor Control Lab

DePauw University | Aug 2023 - Present

- Lead research on motor control, Parkinson's disease, and clinical movement outcomes; design IRB-governed studies and supervise undergraduate researchers.
- Develop statistical analyses, manuscripts, and grant applications; teach Biomechanics, Motor Control, Sport and Exercise Psychology, Sports Analytics, and first-year interdisciplinary coursework.

Data Science Intern

Chinese American Biopharmaceutical Society, Remote | Jun 2026 - Sep 2026

- Built OpenTrial, integrating public clinical and regulatory evidence with Bayesian trial-design calculations, assurance analysis, and simulation.

Postdoctoral Research Fellow

MGH Institute of Health Professions, Brain Recovery Lab | Jun 2022 - Jun 2023

- Analyzed longitudinal fMRI and TMS data from NIH-funded neurological studies using multilevel models and reproducible preprocessing workflows; mentored undergraduate researchers.

Graduate Research and Teaching Assistant

Louisiana State University | Aug 2017 - May 2022

- Conducted studies of Parkinson's disease, visuomotor adaptation, EEG, motor learning, and human movement; taught and guest lectured in biomechanics, statistics, and research methods.

EARLIER APPOINTMENTS

Mitacs Accelerate Research Intern

Memorial University, Faculty of Medicine | Dec 2015 - Jul 2016

Evaluated a tele-ultrasound system for remote clinical use in a feasibility study supported by $10,000 CAD in competitive funding; co-authored a peer-reviewed publication.

Teaching Assistant and SPSS Instructor

University of Ghana | Sep 2012 - Apr 2014

Led tutorials and quantitative-methods instruction in research methods, psychostatistics, community psychology, and SPSS.

RESEARCH AGENDA AND METHODS

Clinical and movement research: Parkinson's disease; motor control and learning; visuomotor adaptation; gait and postural control; neurological rehabilitation; clinical measurement.

Statistical methods: Bayesian inference; multilevel and mixed-effects models; longitudinal and repeated-measures analysis; causal inference; predictive modeling; experimental design; sensitivity analysis.

Computation and data: Python, R, SQL, MATLAB, PyMC, PyTorch, scikit-learn, XGBoost; signal processing, computer vision, and reproducible pipelines; EEG, ECG, fMRI, TMS, motion capture, and behavioral outcomes.

PEER REVIEWED PUBLICATIONS

Addison, R. N., & Van Gemmert, A. W. (2023). Bilateral transfer of a visuomotor task in different workspace configurations. Journal of Motor Behavior. doi:10.1080/00222895.2023.2293002.

Smith, A., Addison, R., Rogers, P., Stone-McLean, J., Boyd, S., Hoover, K., Pollard, M., Dubrowski, A., & Parsons, M. (2018). Remote mentoring of point-of-care ultrasound skills to inexperienced operators using multiple telemedicine platforms: Is a cell phone good enough? Journal of Ultrasound in Medicine, 37(11), 2517-2525. doi:10.1002/jum.14609.

SUBMITTED MANUSCRIPTS

Addison, R. N., et al. (2026). A Parsimonious Predictive Model for Hypertension in African American Adults in the Jackson Heart Study. American Journal of Hypertension. Under review. Manuscript AJH-D-26-00235.

Hondzinski, J., et al., including Addison, R. N. (2026). Evidence for preserved gait consistency after wild blueberry ingestion by individuals with Parkinson's disease. Submitted to Journal of Parkinson's Disease. Manuscript PKN-26-0537.

SELECTED CONFERENCE PRESENTATIONS

Gauss, T. M., Addison, R. N., et al. (2025). Does wild blueberry supplementation improve motor symptoms and performance in people with Parkinson's disease? Society for Neuroscience.

Addison, R. N., Steinberg, F., & Van Gemmert, A. W. (2024). Neural activity associated with visuomotor adaptation in various workspace locations. NASPSPA.

Gauss, T. M., Addison, R. N., et al. (2024/2025). Subtyping Parkinson's disease by symptom dominance reveals group differences in postural sway. NASPSPA.

Wang, Y., Hu, D., Addison, R. N., et al. (2023). Functional changes in superior temporal gyrus in focal dystonia. 6th International Dystonia Symposium.

Addison, R. N., Hondzinski, J. M., & Van Gemmert, A. W. (2022). Training visuomotor adaptation with remembered targets. NASPSPA.

SELECTED COMPUTATIONAL RESEARCH PROJECTS

OpenTrial

CABS | Bayesian clinical trial design | 2026

Built source-cited reports that connect clinical evidence to evidence-derived priors, power and assurance grids, prior sensitivity, and simulation-calibrated group-sequential designs. Integrates ClinicalTrials.gov, PubMed, openFDA, and DailyMed. Research and learning tool, not a validated clinical application. github.com/reubebe/OpenTrial

KinetiScan

Independent | Markerless 3D movement analysis | Ongoing

Developing a local system for recovering joint kinematics and gait metrics from video using computer vision, PyTorch, and ONNX. Distinguishes evaluation of the kinematics engine from the prospective end-to-end validation still needed for camera-based measurement.

ECG Signal Quality Classification

Georgia Tech analytics practicum | Python and PyTorch | 2026

Compared five classifiers on PTB-XL, including logistic baselines and a CNN-Transformer ensemble. Included threshold optimization, calibration assessment, saliency analysis, and lead-level interpretation. github.com/reubebe/ECG-signal-quality-detection

TEACHING AND CURRICULAR DEVELOPMENT

DePauw University: Motor Control; Biomechanics and Lab; Sport and Exercise Psychology; Sports Analytics; First-Year Seminar in Discover Human Performance.

Louisiana State University: graduate teaching and guest instruction in motor learning, biomechanics, statistics, and research methods. University of Ghana: tutorials in research methods, psychostatistics, community psychology, and SPSS.

- Developed Sports Analytics coursework incorporating statistics, predictive modeling, visualization, and applied decision-making.
- Expanded Biomechanics instruction with algebra review, applied labs, and motion-analysis technology; added concept recaps, application examples, assignment rubrics, and structured feedback.

GRANTS AND RECOGNITION

HHMI Departmental Project, DePauw University: co-developed a funded curriculum and space initiative, $15,000 (2024).

Mitacs Accelerate Research Project: $10,000 CAD (2015-2016).

American Kinesiology Association Scholar Award (2021).

CHSE Dean's GA TIGER Travel Grant, LSU (2018-2020; $1,000 each); Graduate Student Association Travel Awards (2018, $250; 2019, $100); Lillian Oleson Scholarship (2018, $1,000).

ACADEMIC AND PROFESSIONAL SERVICE

Founding Faculty Supporter, DePauw MEDLIFE Chapter (2024-Present): advised student leaders on recognition, recruitment, and connections between global health, service, and academic programs.

DePauw service: Kinesiology tenure-track search committee; departmental diversity, equity, and inclusion work; Science Research Fellows involvement; first-year advising and teaching.

Reviewer, Human Movement Science (2023). Membership Committee volunteer, Midwest Chapter of the American College of Sports Medicine.

Professional memberships: NASPSPA and Society for Neuroscience, since 2018.